Project Title: *"Predicting Rainfall Using Machine Learning Classification Algorithms"*

Group Members: Ayesha Ali (23k-7317) and Tehreem Khan (23k-7307)

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Instructors Name: Sir Farukh and Sir Adil

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Introduction:

Rainfall prediction is one of the most complex and uncertain tasks in meteorology, yet it has a profound impact on human society. Timely and accurate rainfall forecasting can proactively help mitigate human and financial losses caused by unexpected weather events. This study aims to develop a machine learning-based model capable of predicting whether it will rain tomorrow based on the weather data from the previous day. The dataset used in this study is sourced from major cities across Australia, offering a diverse and representative set of weather conditions.

As an enthusiast of meteorological predictions, I was intrigued by the parameters that experts use to make weather forecasts. Despite its apparent simplicity, this dataset presents an excellent starting point for exploring machine learning techniques in the domain of weather prediction. By the end of this study, we will explore the following key aspects of the rainfall prediction task:

- Handling Imbalanced Datasets:

Addressed the imbalance between "No Rain" and "Rain" days by applying techniques to balance the data, ensuring the model doesn’t favor the majority class.

- Converting Categorical Data to Numbers:

Transformed categorical features like 'WindGustDir' (weather conditions) into numerical values to make the data more understandable for the model.

- Handling Missing Data:

Used Multiple Imputation by Chained Equations (MICE) to accurately fill in missing values in the dataset, ensuring a complete and reliable dataset.

- Removing Outliers:

Identified and removed extreme values (outliers) that could distort model predictions, ensuring the data was clean and representative.

- Feature Selection:

Applied feature selection techniques to choose the most important features, improving the model's performance by making it faster and more efficient.

- Algorithm Testing and Comparison:

Tested 5 different algorithms on both unbalanced and balanced datasets, and compared their performance to select the most suitable algorithm for the prediction task.

Problem:

We are trying to predict if it will rain tomorrow based on weather data from Australia. This prediction would help in forecasting weather worldwide and take precautionary measures accordingly.

Literature Review:

Rainfall prediction is a critical research area, particularly in regions like Australia with diverse climatic conditions and significant reliance on water resources. Previous researches have emphasized the limitations of traditional statistical models, citing their inability to capture nonlinear relationships in meteorological data. Instead, machine learning algorithms, such as Random Forest, Gradient Boosting, and Neural Networks, are highlighted as superior due to their robustness in handling complex datasets and interactions between variables. Similarly, other research paper, builds upon this foundation, incorporating Neural Networks alongside Random Forest and k-Nearest Neighbors, demonstrating their efficiency in leveraging large-scale data to improve prediction accuracy. Our research paper expands on these approaches by incorporating these algorithms alongside the addition of the Logistic Regression model, enhancing the comparative analysis and providing further insights into predictive performance.

Methodology:

Dataset Description:

This dataset contains about 10 years of daily weather observations from many locations across Australia. Rain Tomorrow is the target variable to predict. It means -- did it rain the next day, Yes or No? This column is Yes if the rain for that day was 1mm or more.

- **Date** is presented in a format yyyy-mm-dd, and **Location** indicates Australian cities where the weather conditions were recorded.
- **MinTemp** and **MaxTemp** stand for minimum and maximum temperature for each day.
- **Temp9am** and **Temp3pm** stand for temperature (degrees Celsius) in the morning and in the evening respectively.
- **Rainfall** is measured in mm, and it is considered to be raining if the value of rainfall indicator is 1 mm or more.
- **Evaporation** is measured in mm.
- **Sunshine** stands for duration of bright sunshine in the 24 hours to midnight, measured in hours.

- **WindGustDir**, **WindGustSpeed**, **WindDir9am**, **WindDir3pm**, **WindSpeed9am**, **WindSpeed3pm** indicate direction (categorical value) and speed of wind (measured in km/h).
- **Humidity9am**, **Humidity3pm** indicates humidity, measured in percent.
- **Pressure9am**, **Pressure3pm** indicate atmospheric pressure, measured in hectopascals.
- **Cloud9am**, **Cloud3pm** seem to indicate cloud level, measured as fraction of sky obscured by cloud.
- **RainToday** indicates whether it is rain today or not.
- **RainTomorrow** consists of target values.

Technique Used:

Solving a classification problem where the goal is to predict whether it will rain tomorrow based on weather data. The dataset includes various weather features such as temperature, humidity, wind speed, and atmospheric pressure, which are used to classify the outcome into two categories: "Rain" or "No Rain." By analyzing the relationships between these variables, we aim to build a predictive model that can accurately forecast rainfall, helping to improve weather prediction systems and decision-making for activities dependent on weather conditions.

1. Handling Imbalanced Data

- **Problem**: The dataset had a lot more instances where it didn't rain (label "No") compared to when it did rain (label "Yes").
- **Solution**: To address this imbalance, we **oversampled** the "Yes" (rain) cases, meaning we added more data points for the minority class (rain), so the model gets enough examples to learn from both classes.

2. Handling Missing Data

- **Problem**: Some columns (like "Evaporation", "Sunshine", etc.) had missing values.
- **Solution**: We used two approaches:
    1. **Mode Imputation**: For categorical columns, we filled missing values with the most common (mode) value in that column.

2. **Multiple Imputation by Chained Equations (MICE)**: For numerical columns, we used a sophisticated technique to estimate and fill the missing values based on other available data.

3. Removing Outliers

- **Problem**: Some data points were extreme and could negatively affect model performance.
- **Solution**: We detected these outliers using the **Interquartile Range (IQR)** method and removed them. This helped ensure that the model doesn't get misled by extreme values that don't represent the general pattern.

4. Feature Selection

- **Problem**: Not all features (columns) in the dataset are important for predicting whether it will rain tomorrow. Some may even make the model more complex without improving accuracy.
- **Solution**: We used two methods:
  1. **Filter Method (Chi-Square)**: This technique helps identify which features are most strongly associated with the target variable (whether it rains or not). We selected the top 10 most important features based on this.
  2. **Wrapper Method (Random Forest)**: We used a machine learning model (Random Forest) to automatically choose the most important features for prediction.

Algorithms Used:

Logistic Regression

What is Logistic Regression?

Overview:
Logistic Regression is a statistical method used for binary classification tasks, where the outcome is limited to two possible values, such as "Rain" or "No Rain."

How it Works:

- It combines multiple input features (e.g., temperature, humidity) into a single score using a linear equation.
- This score is transformed using the sigmoid function to produce a probability between 0 and 1.

- If the probability exceeds 0.5, the prediction is classified as "Rain"; otherwise, it is "No Rain."

Why Use Logistic Regression:

- It is simple and interpretable.
- It provides insights into how each feature influences the outcome.
- Efficient for solving binary classification problems.

Random Forest

Overview:
Random Forest is an ensemble learning technique that combines the predictions of multiple decision trees to enhance accuracy and reduce overfitting.

How it Works:

- It constructs several decision trees, each trained on a random subset of the data.
- Each tree makes a prediction, and the final result is determined by majority voting among the trees.

Why Use Random Forest:

- It handles complex interactions between features effectively.
- It is robust against overfitting due to its ensemble nature.
- Suitable for datasets with non-linear relationships.

Neural Networks

Overview:
Neural Networks are advanced machine learning models inspired by the structure of the human brain, capable of learning complex patterns in data.

How it Works:

- Data is passed through layers of interconnected nodes, or neurons.
- Each neuron processes a small part of the data, identifying patterns at various levels.
- The output of one layer becomes the input for the next, culminating in a final prediction.

Why Use Neural Networks:

- They excel in capturing subtle, non-linear relationships in large datasets.
- They are highly flexible and adaptable to a wide range of problems.
- Effective for high-dimensional and complex data.

K-Nearest Neighbors

Overview:
KNN is a simple, instance-based learning algorithm that classifies data points based on their similarity to previously seen data.

How it Works:

- It identifies the "k" most similar data points (neighbors) in the dataset.
- The prediction is based on the majority class among these neighbors.

Why Use KNN:

- It is easy to understand and implement.
- Performs well when patterns in the data are distinct.
- Requires minimal assumptions about the data.

Decision Trees

Overview:
A Decision Tree is a flowchart-like model that makes predictions by splitting data based on feature thresholds.

How it Works:

- At each node, the data is split based on a feature and a threshold value.
- This process continues until a leaf node is reached, representing the final prediction.

Why Use Decision Trees:

- They are intuitive and easy to visualize.
- Provide a clear explanation of how decisions are made.
- Work well for datasets with non-linear decision boundaries.

Evaluation Metrics:

1. Accuracy

Accuracy tells us how often the model is correct. It's the ratio of correctly predicted instances (both classes) to the total instances.

Imbalanced Dataset: For most models (Logistic Regression, Decision Tree, Random Forest, ANN), accuracy is extremely high (close to 1.00), which means these models predict the majority class (0 - no rain) very well, but they don't capture the minority class (1 - rain).

Balanced Dataset: Accuracy is lower compared to the imbalanced dataset because both classes are now equally represented, making it a tougher problem for the model. Random Forest has the highest accuracy (around 96%), while Logistic Regression achieves about 79%.

2. Precision

Precision measures how many of the positive predictions (e.g., predicting it will rain) were actually correct.

Imbalanced Dataset: All models achieve perfect precision (1.00) for predicting both classes. This indicates that when the models predict a class (rain or no rain), they are extremely accurate.

Balanced Dataset: Precision drops but is still good for most models. For example, Random Forest has a precision of 0.93 for predicting rain, which means 93% of the time when the model predicts it will rain, it’s correct. Logistic Regression’s precision is 0.76, meaning it’s less accurate when predicting rain.

3. Recall

Recall (also known as sensitivity) measures how well the model identifies all the actual positive instances (e.g., how well the model predicts rain when it actually rains).

Imbalanced Dataset: Again, perfect recall for most models, meaning the models detect almost all instances of rain (class 1) and no rain (class 0). However, this could be due to the model’s tendency to always predict the majority class, leading to perfect recall for the majority class and near-perfect recall for the minority class.

Balanced Dataset: Recall is lower but more balanced. Random Forest, for example, has a recall of 0.96 for predicting rain, meaning it correctly identifies 96% of actual rainy days. Logistic Regression’s recall is lower (0.68), indicating that it misses about 32% of the rainy days.

4. F1-Score

F1-Score is the harmonic mean of precision and recall. It gives a better sense of the balance between precision and recall, especially when the class distribution is imbalanced.

Imbalanced Dataset: Most models again achieve perfect F1-scores (1.00), as precision and recall are both high. This can be misleading in imbalanced data since the model may be predicting the majority class (0 - no rain) extremely well, making it look like it's performing well on both classes.

Balanced Dataset: F1-scores give us a better sense of model performance here. Random Forest has an F1-score of 0.95, showing a good balance between precision and recall. Logistic Regression, on the other hand, has an F1-score of 0.71, indicating room for improvement, particularly in detecting rainy days.

5. Confusion Matrix

The Confusion Matrix provides a breakdown of actual vs predicted classifications:

True Positive (TP): Correctly predicted rain.

True Negative (TN): Correctly predicted no rain.

False Positive (FP): Predicted rain when it didn't rain.

False Negative (FN): Predicted no rain when it did rain.

Imbalanced Dataset: In Logistic Regression, Decision Tree, Random Forest, and ANN, the confusion matrix shows a perfect or nearly perfect result. For example, Random Forest perfectly predicts no rain (TN) and rain (TP) with no false positives or negatives.

Balanced Dataset: The confusion matrix becomes more informative. For example, in Logistic Regression, there are 3,363 false positives (predicted rain when it didn't rain), and 4,882 false negatives (predicted no rain when it rained). This suggests that the model struggles more in predicting rain (minority class).

Results:

1. Balanced Dataset:

In a balanced dataset, the classes are approximately equal, meaning there is an equal number of samples for each class. This provides a good scenario for evaluating how well models can distinguish between both classes (e.g., predicting "RainTomorrow").

- **Random Forest** stands out with the highest performance in a balanced dataset. The precision, recall, F1-score, and accuracy are all very high, indicating that this model is highly effective at predicting both the positive and negative classes. The balanced dataset allows it to perform optimally because there is enough data for both classes to learn from.

- **ANN (Artificial Neural Network)** and **Logistic Regression** also perform well, but their metrics are slightly lower than Random Forest. Logistic Regression, while performing well in general, struggles a bit more with precision for the minority class compared to Random Forest.
- **Decision Tree** and **KNN (K-Nearest Neighbors)** also show decent performance, but they do not perform as well as Random Forest. KNN, in particular, has a slightly lower recall, indicating that it misses a few positive cases.

2. Imbalanced Dataset:

In an imbalanced dataset, one class has many more samples than the other. This makes it difficult for models to correctly predict the minority class because they may be biased toward the majority class.

- **Logistic Regression**, **Decision Tree**, **Random Forest**, and **ANN** perform exceptionally well in the imbalanced dataset. These models achieve nearly perfect precision and recall (close to 1.00), suggesting that they handle the imbalance effectively, likely by learning from the minority class properly.
- **KNN** performs less well in the imbalanced scenario. It shows a significant drop in recall for the minority class, indicating that it has difficulty identifying the minority class as effectively as the other models.

Why Certain Models Perform Better:

- **Random Forest** and **ANN** perform well in both balanced and imbalanced datasets because of their ability to handle complex patterns in data. Random Forest uses multiple decision trees and averages their predictions, while ANN (Neural Networks) is capable of capturing complex relationships in the data.
- **Logistic Regression** and **Decision Trees** also perform well, but they tend to be more sensitive to class imbalance. Logistic Regression, for example, tends to predict the majority class more often when faced with imbalanced data.
- **KNN** struggles with imbalanced datasets because it relies heavily on the proximity of neighbors, which can lead it to predict the majority class for most samples, especially when the minority class is underrepresented.

Contributions

This research represents a significant application of concepts learned during our machine learning coursework, demonstrating the practical implementation of theoretical principles in a real-world problem. By systematically applying and evaluating multiple machine learning models, we have deepened our understanding of algorithm optimization, including data preprocessing, feature selection, taking and comparing model accuracy. Our study highlights the effectiveness of multiple models like Decsion Tree, KNN (add the remaining), and logistic regression in managing diverse meteorological variables. This work not only reinforces our knowledge of machine learning but also contributes a scalable, data-driven framework for rainfall prediction, showcasing how academic learnings can translate into impactful, real-world applications.

Significance

Accurate rainfall prediction is crucial for sectors like agriculture, disaster management, and water resource planning. The findings contribute to improved decision-making and resource allocation by providing robust models capable of predicting rainfall across diverse climatic zones. Additionally, these predictive models can play a pivotal role in addressing urban challenges, such as mitigating smog formation. By predicting rainfall accurately, authorities can anticipate natural cleansing effects of precipitation on air quality, helping to implement timely measures to reduce pollution levels and protect public health in smog-prone areas.